

INTRODUCTION TO COMPUTER SYSTEM, NETWORKS AND SECURITY

Assignment 1 part 1 - Numbers, Encodings and Boolean logic (Reflection)

Submitted by;
Rahul Vedant Pemsing
36305529
rpem0003@student.monash.edu

School of Information Technology
FIT1047

TASK	BRIEF DESCRIPTION
Reflection	Reflect on what was done during the three weeks of classes; what you learned, what was difficult...

FIT1047 Weekly Reflection

Week 1:- Overview, Number systems, Basic Boolean logic

Key Learnings:

- What I consider the most important thing I learned this week is Float point representation IEEE 754 standard. For my A-Levels, I did float point representation but it was different, so learning the IEEE 754 standard was interesting.
- The overview presented by the lecture was also as important as I was able to understand how this module will be taught and accessed.

Personal Insights:

- Float point IEEE 754 is a good way to represent numbers with decimal, but sometimes, it cannot represent those numbers, and just round those numbers, so it isn't a good way to represent numbers relating to money.

Application of Knowledge:

- I used what I learned during week 1 in applied class where we were doing questions on conversion.

Task 3.2: Build a table of Number Systems consisting of Base 10, Base 8, Base 16 & Base 2

□ This table will contain 16 numbers only (decimal numbers 0-15)

□ Represent Base-2 numbers using 4 bits.

□ Use appropriate notations/subscripts to differentiate between different number systems:
 $10_{10} = 12_8 = A_{16} = 1010_2$

Base 10	Base 8	Base 16	Base 2
0 ₁₀	0 ₈	0 ₁₆	0000 ₂
1 ₁₀	1 ₈	1 ₁₆	0001 ₂
2 ₁₀	2 ₈	2 ₁₆	0010 ₂
3 ₁₀	3 ₈	3 ₁₆	0011 ₂
4 ₁₀	4 ₈	4 ₁₆	0100 ₂
5 ₁₀	5 ₈	5 ₁₆	0101 ₂
6 ₁₀	6 ₈	6 ₁₆	0110 ₂
7 ₁₀	7 ₈	7 ₁₆	0111 ₂
8 ₁₀	10 ₈	8 ₁₆	1000 ₂
9 ₁₀	11 ₈	9 ₁₆	1001 ₂
10 ₁₀	12 ₈	A ₁₆	1010 ₂
11 ₁₀	13 ₈	B ₁₆	1011 ₂
12 ₁₀	14 ₈	C ₁₆	1100 ₂
13 ₁₀	15 ₈	D ₁₆	1101 ₂
14 ₁₀	16 ₈	E ₁₆	1110 ₂
15 ₁₀	17 ₈	F ₁₆	1111 ₂

Activity 4: Number System Conversions

Task 4.1: Binary to Decimal and Decimal to Binary

(i) Convert Decimal number 163_{10} to Binary number $\rightarrow 10100101_2$

(ii) Convert Binary number 11000101_2 to Decimal number $\rightarrow 197_{10}$

Task 4.2: Binary to Hexadecimal and Hexadecimal to Binary

(i) Convert Binary number 11101010_2 to Hexadecimal number $\rightarrow EA_{16}$

(ii) Convert Hexadecimal number $FA01_{16}$ to Binary number $\rightarrow 1111101000000001_2$

(iii) Convert Binary number 111000_2 to Hexadecimal number $\rightarrow 38_{16}$

Task 4.3: Hexadecimal to Decimal and Decimal to Hexadecimal

(i) Convert Hexadecimal number $1AF_{16}$ to Decimal number $\rightarrow (1 \times 256) + (A \times 16) + (F \times 1) = 256 + 160 + 15 = 431_{10}$

(ii) Convert Decimal number 151_{10} to Hexadecimal number $\rightarrow (1 \times 256) + (5 \times 16) + (1 \times 1) = 256 + 80 + 1 = 337_{10} = 00000101000101_2$

Floating Points Number Conversion

Multiplication method

$0.25_{10} = 2 \times 10^{-1} + 5 \times 10^{-2}$

Base 10

$0.25 \times 10 = 2.50 \rightarrow 2$

$0.50 \times 10 = 5.00 \rightarrow 5$

Round Down

$25_{10} = 2^4 + 2^1 + 2^0 = 11001_2$

$0.25_{10} = 2^{-2} = 0.01_2$

Base 2

$0.25 \times 2 = 0.50 \rightarrow 0$

$0.50 \times 2 = 1.00 \rightarrow 1$

Round Down

Convert decimal 25.25 to binary:

$25.25_{10} = 25 + 0.25$

$= 11001 + 0.01$

$= 11001.01_2$

$= 2^4 + 2^1 + 2^0 + 2^{-2}$

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Challenges Faced:

- I had difficulty adapting to float point representation IEEE 754, since during my A-Level, we did float point representation but not the IEEE 754 standard, and the difference between the two is very minimal, but still enough to throw me off the wrong method and get the wrong answer.

Future Goals:

- I feel like I have a good grasp of the concepts covered this week. I will continue to practise some problems and improve myself. I have also started reading the materials for week 2, to understand what will be required of me.

Week 2:- Gates and circuits, Karnaugh maps

Key Learnings:

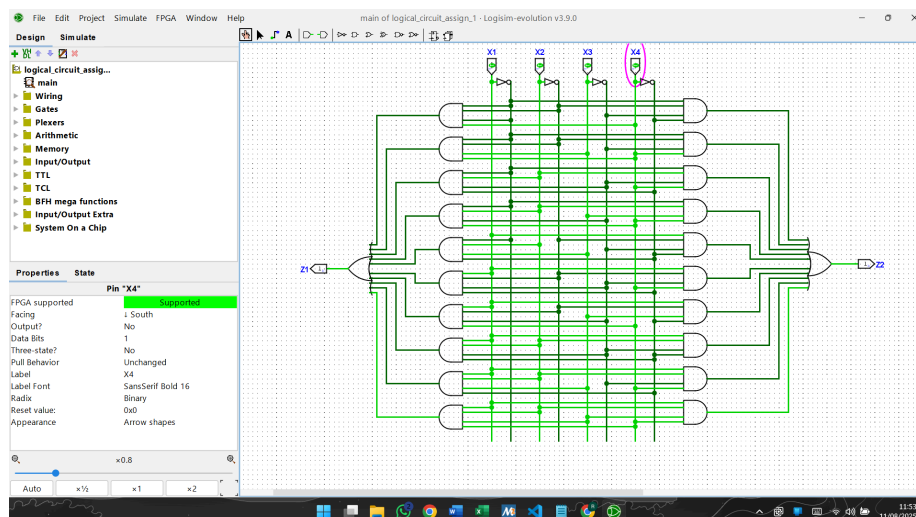
- Logisim; I have never used this software before. This software is a very important tool that will help with building logic circuits.

Personal Insights:

- Learning that the computers are made up of these gates is astonishing, since these are the things that are allowing me to type this reflection.

Application of Knowledge:

- I applied my knowledge when doing my assignment 1 part 2;



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Challenges Faced:

- I am already familiar with gates, circuits and Karnaugh maps. So the most difficult part was getting used to the Logisim software.

Future Goals:

- I plan to use Logisim a lot to be able to get used to it as well as reading the slides for next week's work. I also notice that we will use a new software (<https://marie.js.org/>), so I will mess around with it to see if I can understand what it does before the actual classes.

Week 3:- CPUs, Machine code

Key Learnings:

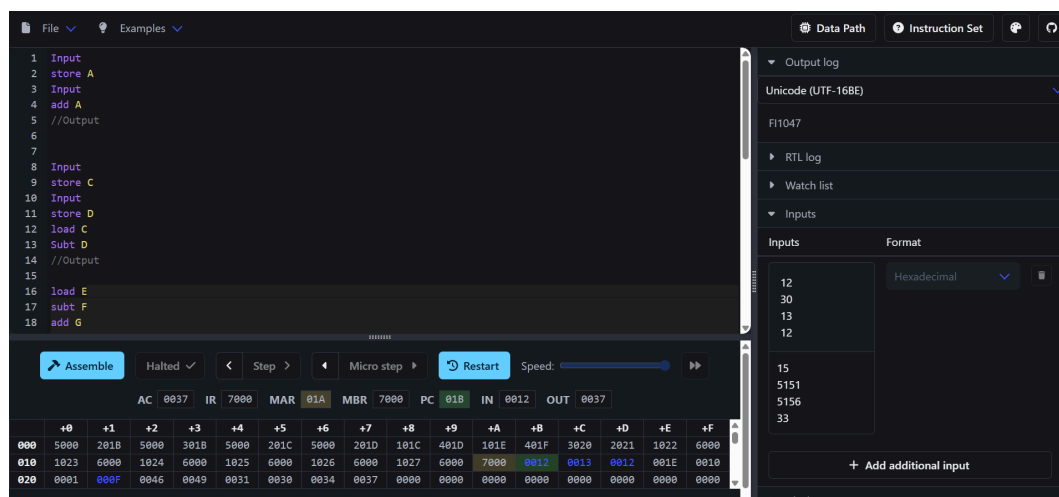
- Understand the Von Neumann architecture
- Write simple assembly code programs using MARIE (loops, multiplication using addition)
- Understand and simulate adder and multiplexer circuits

Personal Insights:

- Every time I hear about how computers work, I am left mesmerised. It is so complex and yet, we use it everyday as if it is something trivial.

Application of Knowledge:

- Coding in assemble:



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Challenges Faced:

- I didn't encounter many challenges this week as I was familiar with the topic from my prior knowledge(A-level) and the indepth pre-reading of the slides before coming to class.

Future Goals:

- I intend to do as I did an in-depth reading of next week' slides, to prepare and be ready for anything to come.